

CarePlus Hospital Web Application Deployment using Docker, Kubernetes (Minikube) and AWS EC2

This document describes the deployment of the CarePlus Hospital web application. The project demonstrates how a static hospital website was containerized using Docker, deployed inside a Kubernetes Pod on Minikube running on an AWS EC2 Ubuntu instance, and exposed through port forwarding.

Project Objectives:

- Design a responsive hospital website.
- Containerize the application using Docker.
- Deploy with Kubernetes Pod.
- Host on AWS EC2 Ubuntu.
- Verify successful deployment using browser access.

Technologies Used:

HTML, CSS, JavaScript, Docker, Dockerfile, Kubernetes, kubectl, Minikube, AWS EC2, Ubuntu 24.04 LTS.

Deployment Steps:

1. Launch Ubuntu EC2 instance.
2. Install Apache (optional), Docker and dependencies.
3. Install kubectl.
4. Install Minikube.
5. Start Minikube using Docker driver.
6. Build Docker image.
7. Create pod.yml manifest.
8. Deploy using kubectl apply.
9. Verify pod status using kubectl get pods.
10. Access application using kubectl port-forward 5000:5000.

Screenshots Included:

Figure 1: Home Page

Figure 2: Services Section

Figure 3: Doctors Section

Figure 4: AWS EC2 Deployment Commands

Results:

The website was successfully deployed and accessible through the EC2 public IP on port 5000. Kubernetes managed the application lifecycle while Docker provided a portable container image.

Conclusion:

This project demonstrates an end-to-end DevOps workflow beginning with application development, containerization, Kubernetes orchestration, and cloud deployment. It provides practical experience with real-world DevOps tools and deployment practices suitable for academic projects and GitHub portfolios.

Figure 1

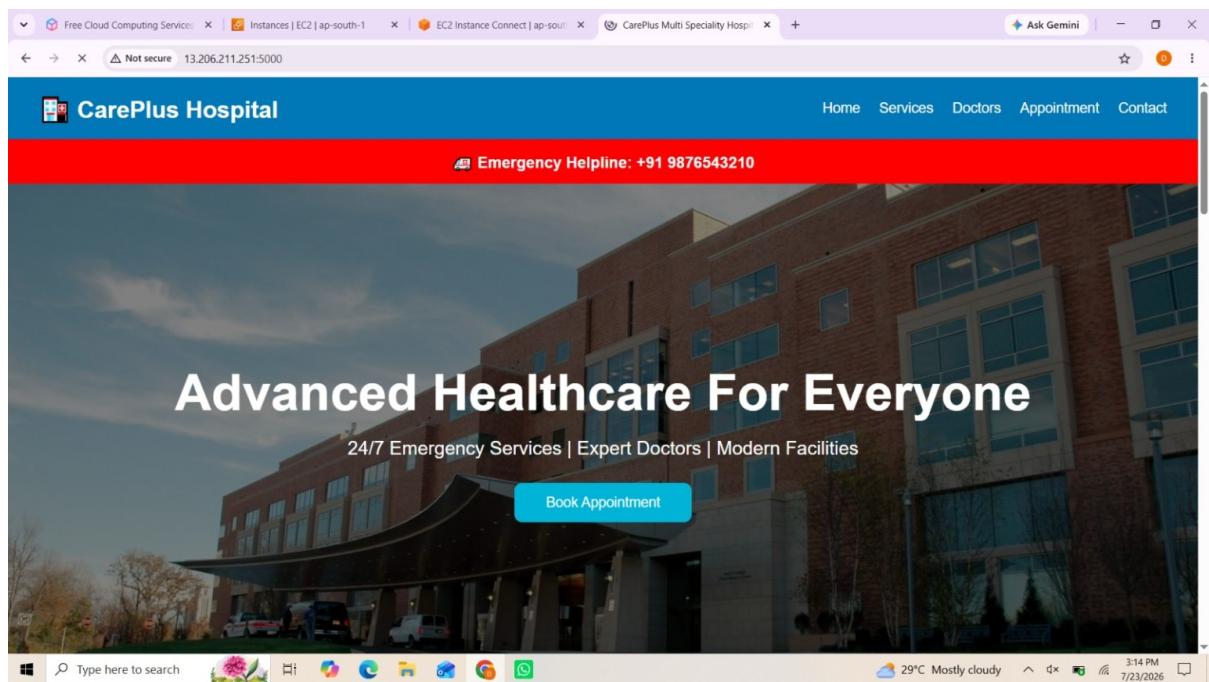


Figure 2

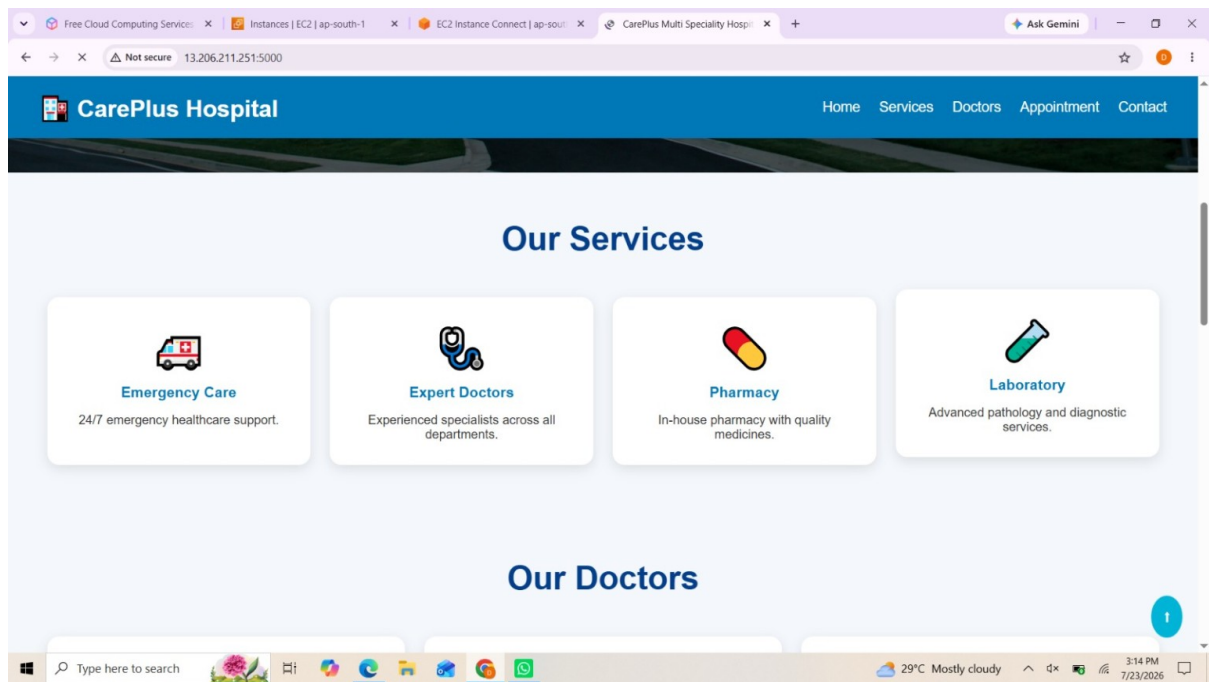


Figure 3

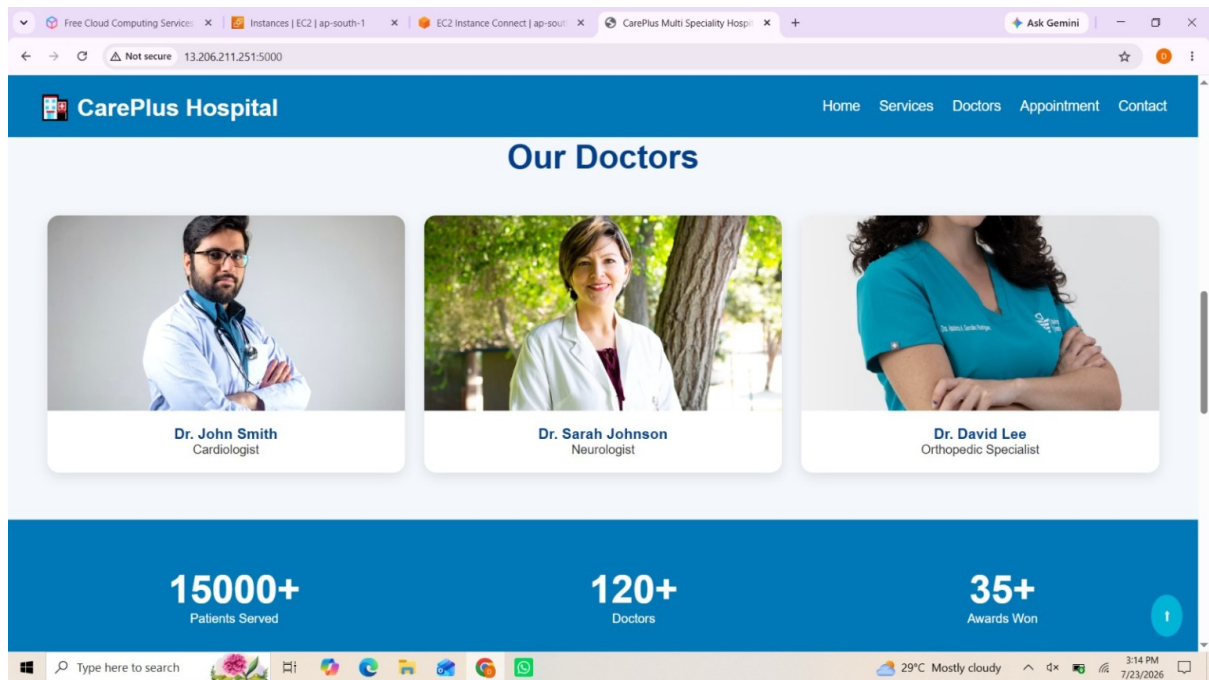


Figure 4

